and hence compute A .

OR

... values and

Total Printed Pages: Roll No. :.....

Paper Code: MAB-101
Examination –May- 2025
B.Tech. I Sem :Common for all branches
Subject Name: Linear Algebra and Calculus

Time : 3Hrs

Max. Marks :60
Min. Marks :19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c &d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a(i) and a(ii) Objective type/Fill in the blanks/true& false.

Part (b) – Three/four sentences based answer question (Maximum 50words)

Part (c) –Short Descriptive Ans./Numerical/ algorithm /programs basedQuestion. (Maximum 100 words)

Part (d) – Long Answer /Shorts -notes /Descriptive Ans. based Question (Maximum 400 words)

Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)- I	When equation of curve is given in the form y equal to FX radius of curvature is given by (i) $\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$ (ii) $\frac{\left[1+\left(\frac{dy}{dx}\right)\right]^{3/2}}{\frac{d^2y}{dx^2}}$ (ii) $\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d\ y}{dx}}$ (iv) *None of these*	01	1	1
	(a) -II	While finding the extreme values of a function z=f(x,y) if rt-s^2>0 and r<0 for a particular solution (a,b) then z has a_____ value at(a,b). (i) maximum (ii) minimum (iii) case is doubtful (iv) none of these	01	1	1
	(b)	Write the Taylor's theorem and its special case Maclaurin's theorem.	02	2	1
	(c)	Expand Sin(x+y) in powers of y using Taylor's theorem and deduce that Sin(x+y)=SinxCosy+SinyCos x.	03	3	1
		OR			
		Find the extreme values of the function f(x)=x^3+ y^3-3axy.	03	3	1
	(d)	In the cycloid $x = a(\theta + sin\theta), y = a(1 - cos\theta)$prove that ρ = 4acos $\frac{1}{2}\theta$.	05	4	1
		OR			
		Obtain the first four term in the expansion of log log sinx in powers of (x– 2).	05	4	1
Q.2	(a)-I	if u is a homogeneous function of degree n in x and y then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} =$ (i) (n-1)u (ii) n(n-1)u (iii) nu (iv) n(u-1)	01	1	2
	(a)-II	Let z=f(x,y) be a function of two independent variables x and y then	01	1	3

how many second order partial derivatives of z will exist

(i) 0 (ii) 1 (iii) 2 (iv) 3

02	2	2

(b) Define homogeneous function. 03 | 4 | 2

(c) If $u = log(x^3 + y^3 + z^3 - 3xyz)$ show that

$$\left(\partial/\partial x + \partial/\partial y + \partial/\partial z\right)^2 u = -\frac{9}{(x+y+z)^2}$$

OR

if $u = \tan^{-1}\frac{x^3+y^3}{x-y}$ prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = Sin2u$ 03 | 4 | 2

(d) A balloon in the form ofright circular cylinder of radius 1.5m and length 4m and is surmounted by hemispherical ends. If the radius is increased by 0.01m and length by 0.05m find the percentage change in the volume of balloon. 05 | 5 | 2

OR

If u=f(r)where $r^2 = x^2 + y^2$ prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + f'(r)/r$ 05 | 5 | 2

Q.3 (a)-I If f(x) is a continuous and single valued function of x in the closed interval [a,b] then 01 | 1 | 3

$$\lim_{h\to 0}[f(a) + f(a+h) + f(a+2h) \pm - - - f(a+\overline{n-1}h)]$$

where nh=b-a is called the

(i) definite integral of f(x) between the limits a and b
(ii) indefinite integral
(iii) definite integral of f(x) between the limits b and a
(iv) None of these

(a)-II Limits of integration are constant then the order of integration is immaterial, is this statement true or false ? 01 | 1 | 3

(b) Define double integral and triple integral. 02 | 2 | 3

(c) Find the limit when n->∞ of the series 1/n+1/(n+1)+1/(n+2)+----- +1/3n. 03 | 3 | 3

OR

Evaluate $\iint e^{2x+3y}dxdy$ over the triangle bounded by x=0, y=0 and x+y=1. 03 | 3 | 3

(d) Evaluate $\iint ydxdy$ over the part of the plane bounded by the line y =x and parabola $y=4x-x^2$. 05 | 4 | 3

OR

Change the order of integration in $\int_0^a \int_0^a \frac{xdxdy}{x^2+y^2}$ and hence evaluate the same. 05 | 4 | 3

Q.4 (a)-I A matrix having only 1 column and any number of rows is known as 01 | 1 | 4

(i) row matrix (ii) column matrix

(iii) augmented matrix (iv) unit matrix

	(a)-II	Which of the following is the method of finding rank (i) transpose method (ii) normal form method (ii) singularization method (iv) none of these	01	1	4
	(b)	State Calley Hamilton theorem.	02	2	4
	(c)	Find the rank of the following matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$	03	3	4
		OR			
		Determine the values of λ and μ such that the system $2x-5y+2z=8$; $2x+4y+6z=5$; $x+2y+\lambda z=\mu$ has (i) no solution (ii) a unique solution (iii) infinite number of solutions.	03	3	4
	(d)	Find the characteristic equation of the matrix $A=\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1}.	1	4	4
		OR			
		Define eigen values and eigen vectors and find the eigen values and eigen vectors of the following matrix $A=\begin{bmatrix} 8 & 6 & -2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	05	4	4
Q.5	(a)-I	In Algebra of propositions $p \vee p=p$ and $p \wedge p=p$ is known as (i) Idempotent law (ii) Commutative law (iii) Associative law (iv) Distributive law	01	1	5
	(a)-II	Let (B,+,-,',0,1) be a Boolean algebra. As the statement obtained by interchanging the binary operations '+' and '.' and interchanging identity elements 0 and 1 in a statement is known as (i) Reciprocal of a statement (ii) inverse of a statement (iii) Dual of any statement (iv) None of these		1	5
	(b)	Define walk path and circuit.	02	2	5
	(c)	Prove that the statement $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction.	03	2	5
		OR			
		State and prove De morgan's laws.	03	2	5
	(d)	In the Boolean algebra (B,+,.,') show that [a'.(a+b)]'+[b.(b+a')]'+[b.(b'+a)]'=1; for all a,b $\in$ B.	05	4	5
		OR			
		Let (B,+,.,') be a Boolean algebra (B,+,.,'). If for a,b,c$\in$B a+b=a+c and a.b=a.c then b=c.	05	4	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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